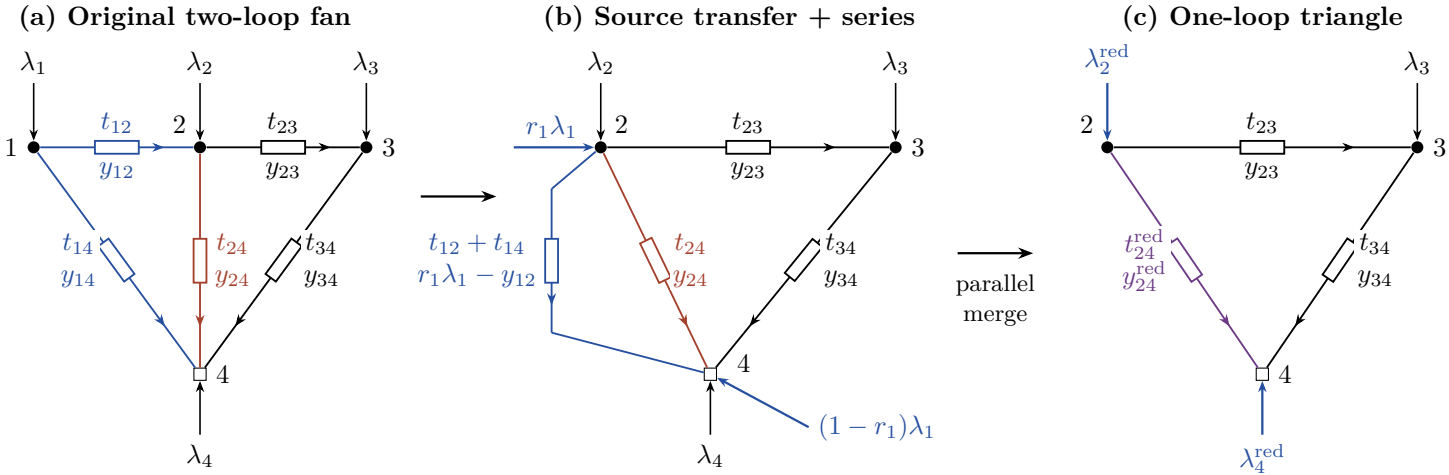


$$r_1 = \frac{t_{14}}{t_{12}+t_{14}}, \qquad r_2 = \frac{t_{12}+t_{14}}{t_{12}+t_{14}+t_{24}}, \qquad \lambda_2^{\mathrm{red}} = \lambda_2 + r_1\lambda_1, \quad \lambda_4^{\mathrm{red}} = \lambda_4 + (1-r_1)\lambda_1.$$

$$-[y_{12} - r_1\lambda_1] = y_{14} - (1-r_1)\lambda_1 = (1-r_2)y_{24}^{\mathrm{red}}, \qquad y_{24} = r_2 y_{24}^{\mathrm{red}}.$$

$$t_{24}^{\mathrm{red}} = \left[\frac{1}{t_{12}+t_{14}} + \frac{1}{t_{24}}\right]^{-1}, \qquad y_{24}^{\mathrm{red}} = \underbrace{r_1\lambda_1 - y_{12}}_{\text{series branch}} + \underbrace{y_{24}}_{\text{original spoke}} = \lambda_2 + r_1\lambda_1 - y_{23}.$$

$$y_{12} = r_1\lambda_1 - (1-r_2)y_{24}^{\mathrm{red}}, \qquad y_{14} = \lambda_1 - y_{12}, \qquad y_{24} = r_2 y_{24}^{\mathrm{red}}.$$